Libro 1

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capitulo 2
problemas 1-45

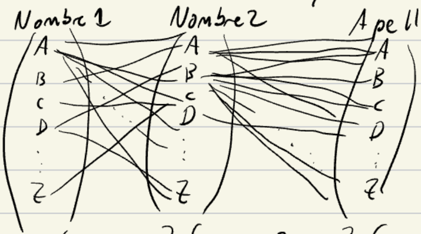
Note: *Assume in each case that all arrangements have the same probability.*

1. How many different sets of initials can be formed if every person has one surname and (*a*) exactly two given names, (*b*) at most two given names, (*c*) at most three given names?

Suponiendo que existen nombres y apellidos de A – Z.

(26 Letras en inglés)

a) Considerando 2 nombres y 1 apellido



$26 \cdot 26 \cdot 26$

$\therefore \exists\ 26^3$ conjuntos diferentes

b) A lo mucho 2 nombres, es 2 nombres y un apellido más 1 nombre y 1 apellido

$a = n_1 + n_2 = 26^3 + 26^2$

c) A lo mucho 3 nombres:

$R = n_3 + n_2 + n_1 = 26^4 + 26^3 + 26^2$

posibles con 3 nombres

2. Letters in the Morse code are formed by a succession of dashes and dots with repetitions permitted. How many letters is it possible to form with ten symbols or less?

3. Each domino piece is marked by two numbers. The pieces are symmetrical

Supongamos que tenemos una secuencia de i dígitos entre '·' y '—'

$$\Rightarrow n_i = 2^i$$

Ahora para 10 dígitos o menos, debemos considerar que $i \in [1, 10]$

$$\Rightarrow n_{i \le 10} = \sum_{i=1}^{10} 2^i$$

3. Each domino piece is marked by two numbers. The pieces are symmetrical so that the number-pair is not ordered. How many different pieces can be made using the numbers 1, 2, ..., n?

4. The numbers 1, 2, ..., n are arranged in random order. Find the proba-

Supongamos un numero $x \in [1, 9]$, una ficha de domino contiene $(x_1, x_2) \in D$, $D \subseteq xRx$, donde $(x_1, x_2) = (x_2, x_1)$ ∴

$$\boxed{x_1 \mid x_2} = \boxed{x_2 \mid x_1}$$

Por lo que debemos considerar una combinación sin remplazo, así como los propios pares de numeros.

$$n = n_{pares} + n_{combinaciones} = 9 + \binom{9}{2} = 9 + \frac{9!}{2!(7)!} = 9 + \frac{9 \cdot 8}{2}$$

$$n = 9 + \frac{72}{2} = 9 + 36 = 45 \text{ fichas distintas}$$

made using the numbers 1, 2, ..., n?

4. The numbers 1, 2, ..., n are arranged in random order. Find the probability that the digits (*a*) 1 and 2, (*b*) 1, 2, and 3, appear as neighbors in the order named.

Supongamos m numeros ordenados de una sola manera, entonces nos quedan $(n-m)!$ formas de ordenar los n numeros. También podemos acomodar los m ordenados entre los $(n-m+1)$ espacios que se dan. $\Rightarrow$ Para $R_n = (n-m)!(n-m+1)$

∴ a) $R_2 = (n-2)!(n-2+1) = (n-2)!(n-1)$ a)

b) $R_3 = (n-3)!(n-3+1) = (n-3)!(n-2)$ b)

5. *A* throws six dice and wins if he scores at least one ace. *B* throws twelve dice and wins if he scores at least two aces. Who has the greater probability to win?[18]

Hint: Calculate the probabilities to lose.

A/ Tener n dados La cantidad de diferentes resultados es

$$N_n = \frac{6^n \leftarrow \# \text{ caras}}{2^n \leftarrow (A,B)=(B,A)}$$

La cantidad de casos favorables es cuando al menos 2 dados tienen el mismo valor.

$$N_{f_n} = \frac{n \cdot 6^{n-2}}{2^{n-2}} \qquad \text{pares de } \{1, 2, \ldots, 6\}$$

Entonces la probabilidad es:

$$P_n = \frac{\frac{n \cdot 6^{n-2}}{2^{n-2}}}{\frac{6^n}{2^n}} = \frac{n \cdot 6^{n-2} \cdot 2^n}{6^n \cdot 2^{n-2}} = n \frac{2^2}{6^2} = n \frac{4}{36} = n \frac{1}{9}$$

∴ Por lo tanto hay mayor probabilidad con 12 dados

$$12 \cdot \frac{1}{9} > 6 \cdot \frac{1}{9}$$

6. (*a*) Find the probability that among three random digits there appear exactly 1, 2, or 3 different ones. (*b*) Do the same for four random digits.

Sea n la cantidad de digitos

$$N_n = n^{10} \quad [0,9] \qquad N_{f_n} = n!$$

a) $n=3 \Rightarrow P_3 = \frac{N_{f_3}}{N_3} = \frac{3!}{3^{10}}$

b) $n=4$ $P_4 = \frac{N_{f_4}}{N_4} = \frac{4!}{4^{10}}$

7. Find the probabilities p_r that in a sample of r random digits no two are equal. Estimate the numerical value of p_{10}, using Stirling's formula.

Para que los digitos sean distintos implica permutacion sin remplazo

$\Rightarrow$ Para r digitos ?

$$N_{f_r} = \prod_{i=r}^{10} i = \frac{10!}{(10-r)!} \qquad N_r = r^{10} \quad \Rightarrow P_r = \frac{10!}{(10-r)!\, r^{10}}$$

$\forall r \in [0,9]$

Luego para $r=10 \Rightarrow P_r = \frac{10!}{(10-10)!\, 10^{10}} = \frac{10!}{10^{10}}$

equal. Estimate the numerical value of p_{10}, using Stirling's formula.

8. What is the probability that among k random digits (*a*) 0 does not appear; (*b*) 1 does not appear; (*c*) neither 0 nor 1 appears; (*d*) at least one of the two digits 0 and 1 does not appear? Let A and B represent the events in (*a*) and (*b*). Express the other events in terms of A and B.

a) $P_{a)} = \dfrac{n \to \text{casos fav.}}{N \to \text{casos totales}}$

$n = 9^k \leftarrow$ porque se excluye al $\emptyset$

$N = 10^k \leftarrow \{0,1,2,3,4,5,6,7,8,9\}$

$\Rightarrow P_{a)} = \dfrac{9^k}{10^k}$ //

b) $P_{b)} = \dfrac{9^k}{10^k}$ // $\leftarrow$ se excluye al 1

c) $P_{c)} = \dfrac{8^k}{10^k}$ // $8^k \leftarrow \{2,3,\ldots,9\}$, $10^k \leftarrow \{0,1,\ldots,8,9\}$

d) $P_{d)} = P_{a)} + P_{b)} - P_{c)} = \dfrac{9^k}{10^k} + \dfrac{9^k}{10^k} - \dfrac{8^k}{10^k}$

$= \dfrac{2(9^k) - 8^k}{10^k}$ //

(*b*). Express the other events in terms of A and B.

9. If n balls are placed at random into n cells, find the probability that exactly one cell remains empty.

10. At a parking lot there are twelve places arranged in a row. A man ob-

Como son n celdas y n posibles pelotas, suponiendo que una celda puede tener $[0, n]$ pelotas.

Supongamos que forzamos la primer celda a tener al menos 1 pelota, entonces tiene n posibilidades de ser llenada, y las siguientes celdas $[0,(n-1)]$ posibilidades,

$\square \quad \square \quad \square \quad \square \quad \cdots \quad \square$

$[1,n] \quad [0,n-1] \quad [0,n-1] \quad [0,n-1] \quad [0,n-1]$

$\Rightarrow N = n^n \leftarrow n$ posibilidades

(n: n celdas)

si forzamos a la celda 1 a no tener pelotas, nos quedan $n-1$ celdas a elegir

$\Rightarrow n = (n-1)^n$

$\therefore P_n = \dfrac{(n-1)^n}{n^n}$ //

exactly one cell remains empty.

10. At a parking lot there are twelve places arranged in a row. A man observed that there were eight cars parked, and that the four empty places were adjacent to each other (formed *one* run). Given that there are four empty places, is this arrangement surprising (indicative of non-randomness)?

$N = 2^k$; $k = 12$ espacios

$N_{fav} = 1$

$\therefore P_k = \dfrac{1}{2^k} = \dfrac{1}{28}$

$\Rightarrow P_k < 1/2 \therefore$ Es poco probable que sea producto del azar

places, is this arrangement surprising (indicative of non-randomness)?

11. A man is given n keys of which only one fits his door. He tries them successively (sampling without replacement). This procedure may require 1, 2, . . . , n trials. Show that each of these n outcomes has probability n^{-1}.

Suponiendo casos totales de intento $N = n$
y casos favorables $N_f = 1$

$$\Rightarrow P_n = \frac{N_f}{N} = \frac{1}{n} = n^{-1}$$

12. Suppose that each of n sticks is broken into one long and one short part. The $2n$ parts are arranged into n pairs from which new sticks are formed. Find the probability (*a*) that the parts will be joined in the original order, (*b*) that all long parts are paired with short parts.[19]

En total resultan $2n$ palos y se hacen n parejas

Sea S el conjunto de palos, tal que, S está particionado por $\mathcal{L}$ e I.

$S = \mathcal{L} \cup I$; $\mathcal{L} \cap I = \phi$

$\mathcal{L} = \{ l : \text{palos largos} \}$

$I = \{ i : \text{palos cortos} \}$

Imaginemos que hacemos los pares de palos de la siguiente manera:

$$(s_1, s_2), (s_3, s_4), (s_5, s_6), \ldots, (s_{2n-1}, s_{2n})$$

Notemos que tener $(s_1, s_2), (s_3, s_4)$ es lo mismo que tener $(s_3, s_4), (s_1, s_2)$; entonces debemos garantizar que el orden de las tuplas no afecten mi conteo.

Supongamos que primero seleccionamos al primer elemento de cada tupla.

$$(s_1, \ )(s_3, \ )(s_5, \ ), \ldots, (s_{2n-1}, \ )$$

Entonces N está dado por la combinatoria.

$$N = \binom{2n}{n}$$

Nos quedan n palos restantes, los cuales crearemos los diversos pares combinándolos con las $n!$ permutaciones.

$$N = \binom{2n}{n} \cdot n!$$

Luego como es lo mismo (a,b) que (b,a) reducimos las 2^n repeticiones

$$N = \binom{2n}{n} \frac{n!}{2^n}$$

$$\Rightarrow N = \binom{2n}{n} = \left(\frac{2n!}{\cancel{n!}(2n-n)!}\right) \cdot \frac{\cancel{n!}}{2^n} = \frac{2n!}{n!\,2^n}$$

a) $P_a = \dfrac{1}{\dfrac{2n!}{n!\,2^n}} = \dfrac{n!\,2^n}{(2n)!}$

b) $P_b = n! \cdot \dfrac{n!\,2^n}{(2n)!} = \dfrac{(n!)^2\,2^n}{(2n)!}$

13. *Testing a statistical hypothesis.* A Cornell professor got a ticket twelve times for illegal overnight parking. All twelve tickets were given either Tuesdays or Thursdays. Find the probability of this event. (Was his renting a garage only for Tuesdays and Thursdays justified?)

Cada semana se da un día de multa: $N = 7$ $\Rightarrow P_2 = \frac{2}{7}$

de la semana es Martes o Jueves: $N_{fav} = 2$

Luego para que se repita 12 veces: $f(12) = \binom{12}{12} p^{12} (1-p)^{12-12}$

$f(12) = \left(\frac{2}{7}\right)^{12}$ ← Binomial

Dado que $\left(\frac{2}{7}\right)^{12} < \frac{1}{2}$ sí lo justificaría.

14. *Continuation.* Of twelve police tickets none was given on Sunday. Is this evidence that no tickets are given on Sundays?

Se requerirían más muestras para poder deducir que en sabados no se dan multas.

15. A box contains ninety good and ten defective screws. If ten screws are used, what is the probability that none is defective?

$P_{good} = \frac{90}{100} = \frac{9}{10} \Rightarrow f(12) = \left(\frac{9}{16}\right)^{12}$

← Binomial

16. From the population of five symbols $a, b, c, d, e,$ a sample of size 25 is taken. Find the probability that the sample will contain five symbols of each

- $N_{fav} = 1 \cdot 5^{20} = 5^{20}$
- $N = 5^{25}$

$\Rightarrow P_{5/25} = \frac{N_{fav}}{N} = \frac{5^{20}}{5^{25}} = \frac{1}{5^5}$

18. What is the probability that two throws with three dice each will show the same configuration if (*a*) the dice are distinguishable, (*b*) they are not?

a) $N_{fav} = 1$ ← una conf.

$N = 3!\, 6^3$ ← dados.

$3!$ ← son distinguibles; 6 ← caras

$P = \frac{1}{3!\, 6^3} = \frac{1}{6 \cdot 6^3} = \frac{1}{6^4}$

Luego $f(2) = \binom{2}{2} \cdot p^2 \cdot (1-p)^{2-2} = p^2 = \frac{1}{6^8}$

b) $N_{fav} = 1$ ← una conf.

$N = 6^3$ ← dados.

6 ← caras

$P = \frac{1}{6^3}$

$\Rightarrow f(2) = \binom{2}{2} \cdot p^2 \cdot (1-p)^{2-2} = p^2 = \frac{1}{6^6}$

19. Show that it is more probable to get at least one ace with four dice than at least one double ace in 24 throws of two dice. The answer is known as de Méré's paradox.[21]

A) Un uno con 4 dados:

$$N_{A\,fav} = 1\cdot \underset{\text{caras}}{6}^{\overset{\text{dados}}{3}} = 6^3 \quad ; \quad N_A = 6^4$$

$$\Rightarrow P_A = \frac{6^3}{6^4} = \frac{1}{6}$$

B) Dobles Uno con 24 tiradas con 2 dados:

$$N_{B\,fav} = 1 \quad ; \quad N_B = \underset{\text{caras}}{6}^{\overset{\text{dados}}{2}}$$

$$\Rightarrow P_B = \frac{1}{6^2}$$

Luego Tenemos N posibles tuplas o pares de dados posibles, de esos pares elegimos 23 casos cuales sean y 1 para asegurar el almenos 1 doble unos, entre los 24 tiros posibles

$$N_{fav(24)} = 1\cdot(N_B)^{23} \quad ; \quad N_{(24)} = (N_B)^{24} \quad \Rightarrow P_{(24)} = \frac{(N_B)^{23}}{(N_B)^{24}} = \frac{1}{N_B} = \frac{1}{6^2}$$

$$\Rightarrow P[24 \wedge B] = P_{(24)} \cdot P_B = \frac{1}{6^2}\cdot\frac{1}{6^2} = \frac{1}{6^4}$$

$$\therefore \text{Como } 6^4 > 6 \Rightarrow \frac{1}{6^4} < \frac{1}{6} \Rightarrow P[24\wedge B] < P_A$$

20. From a population of n elements a sample of size r is taken. Find the probability that none of N prescribed elements will be included in the sample, assuming the sampling to be (a) without, (b) with replacement. Compare the numerical values for the two methods when (i) $n = 100$, $r = N = 3$, and (ii) $n = 100$, $r = N = 10$.

a) $$\underset{\text{totales}}{T} = \frac{n!}{(n-r)!} \quad ; \quad \underset{\text{sin } N\text{'s}}{T_{(n-N)}} = \frac{(n-N)!}{(n-N-r)!}$$

$$P^c_{(n-N)} = \frac{T_{(n-N)}}{T} = \frac{\frac{(n-N)!}{(n-N-r)!}}{\frac{n!}{(n-r)!}} = \frac{(n-N)!\,(n-r)!}{n!\,(n-N-r)!}$$

b) $$T = n^r \quad T_{(n-N)} = (n-N)^r \quad P^s_{(n-N)} = \frac{T_{(n-N)}}{T} = \frac{(n-N)^r}{n^r}$$

i) $$P^c_{(n-N)} = \frac{(100-3)!\,(100-3)!}{100!\,(100-3-3)!} = \frac{((100-3)!)^2}{100!\,(100-6)!} \qquad P^s_{(n-N)} = \frac{(100-3)^3}{100^3}$$

ii) $$P^c_{(n-N)} = \frac{[(100-10)]^2}{100!\,(100-20)!} \qquad P^s_{(n-N)} = \frac{(100-10)^{10}}{100^{10}}$$

22. *Chain letters*. In a population of $n + 1$ people a man, the "progenitor," sends out letters to two distinct persons, the "first generation." These repeat the performance and, generally, for each letter received the recipient sends out two letters to two persons chosen at random without regard to the past development. Find the probability that the generations number $1, 2, \ldots, r$ will not include the progenitor. Find the median of the distribution, supposing n to be large.

Total de elegir 2 personas de n : $\binom{n}{2} = \frac{n(n-1)}{2}$

Total de no elegir a p: $n-1$

$$\Rightarrow P_{env} = \frac{\cancel{n-1}}{\frac{n\cancel{(n-1)}}{2}} = \frac{2}{n} \Rightarrow q_{env} = \left(1-\frac{2}{n}\right)$$

Numero de envios por generación: $K = 2^1 + 2^2 + \cdots + 2^{r-1} = 2^r - 2$

$$\Rightarrow P_{(sin\ p)} = \left(1-\frac{2}{n}\right)^{2^r-2}$$

$$E[X] = \sum_{x=1}^{n} (x)\left(1-\frac{2}{x}\right)^{2^r-2}$$

23. *A family problem*. In a certain family four girls take turns at washing dishes. Out of a total of four breakages, three were caused by the youngest girl, and she was thereafter called clumsy. Was she justified in attributing the frequency of her breakages to chance? Discuss the connection with random placements of balls.

Supongamos $p = 1/4$ de romper platos

$$\Rightarrow P(X=3) = \binom{4}{3} p^3 (1-p)^1 = \frac{4!}{3!(1)!} \left(\frac{1}{4}\right)^3 \cdot \left(\frac{3}{4}\right) = 4 \cdot \left(\frac{1}{64}\right) \cdot \left(\frac{3}{4}\right) = \frac{3}{64} \approx 0.0468$$

∴ De los 4 experimentos es poco probable que si rompa 3 platos, por lo que si es justificable XD.

24. What is the probability that (*a*) the birthdays of twelve people will fall in twelve different calendar months (assume equal probabilities for the twelve months), (*b*) the birthdays of six people will fall in exactly two calendar months?

a) $N = 12^{12}$ $N_{fav} = 12! \Rightarrow P_{12} = \frac{12!}{12^{12}}$

b) $N = 12^6$ $N_{fav} = \binom{12}{2} = \frac{12!}{2!(10)!} = \frac{12 \cdot 11}{2} = 6 \cdot 11 = 66 \Rightarrow P_6 = \frac{66}{12^6}$

25. Given thirty people, find the probability that among the twelve months there are six containing two birthdays and six containing three.

$N = 12^{30}$

$N_{fav} = 2^6 + 3^6$ $\Rightarrow P_n = \frac{2^6 + 3^6}{12^{30}}$

26. A closet contains n pairs of shoes. If $2r$ shoes are chosen at random (with $2r < n$), what is the probability that there will be (*a*) no complete pair, (*b*) exactly one complete pair, (*c*) exactly two complete pairs among them?

Tenemos pues n Izquierdos y n derechos $\Rightarrow N = \frac{2n!}{n!\,2^n}$

a) Si seleccionamos un par, tenemos n-2 zapatos a elegir para que sean Incompatibles.

$\Rightarrow N_a = \binom{n-2r}{r} = \frac{(n-2r)!}{r!\,(n-r)!}$ $\quad P_a = \frac{(n-2r)!\,n!\,2^n}{r!\,(n-r)!\,(2n)!}$

b) $N_b = \binom{2n-2r-2}{2r-2} = \frac{2(n-r-1)!}{2(r-1)!\,2(n-2r)!} = \frac{(n-r-1)!}{(r-1)!\,2(n-2r)!} = \frac{1}{2}\frac{(n-r-1)!}{(r-1)!(n-2r)!}$ $\quad P_b = \frac{1\,(n-r-1)!\,n!\,2^n}{2(r-1)!\,(n-2r)!\,(2n)!}$

c) $N_c = \binom{2n-2r-4}{2r-4} = \frac{(2n-2r-4)!}{(2r-4)!\,(2n-4r)!} = \frac{(n-r-2)!}{(r-2)!(n-2r)!} \Rightarrow P_b = \frac{(n-r-2)!\,n!\,2^n}{(r-2)!\,(n-2r)!\,(2n)!}$

28. A group of $2N$ boys and $2N$ girls is divided into two equal groups. Find the probability p that each group will be equally divided into boys and girls. Estimate p, using Stirling's formula.

$$p = \frac{(2N)!\cdot(2N)!}{(4N)!} = \frac{(2N)!^2}{(4N)!}$$

31. What is the probability that the bridge hands of North and South together contain exactly k aces, where $k = 0, 1, 2, 3, 4$?

Total de cartas: 52 Total de Aces: 4 Cartas por mano: 13

Sea $K \in [1,2] \Rightarrow$

Casos Posibles: $N = \underbrace{\binom{52}{13}}_{\text{Norte}}\underbrace{\binom{39}{13}}_{\text{Sur}}$ $\quad$ (52-13)

Casos Exitosos: $n_A = \underbrace{\binom{52-K}{13-K}}_{\text{Norte}}\underbrace{\binom{39-2K}{13-K}}_{\text{Sur}}$ $\quad \therefore P_{n_A} = \frac{\binom{52-K}{13-K}\binom{39-2K}{13-K}}{\binom{52}{13}\binom{39}{13}}$ $\forall K \leq 2$

Para $K \in [3,4]$ como solo existen 4 Aces y K representa la cantidad que obtiene Norte y Sur, entonces no hay casos en los que el Norte y el Sur obtengan K Aces a la vez.

$\therefore P_{n_A} = 0 \quad \forall K \geq 3$

32. Let a, b, c, d be four non-negative integers such that $a+b+c+d = 13$. Find the probability $p(a, b, c, d)$ that in a bridge game the players North, East, South, West have a, b, c, d spades, respectively. Formulate a scheme of placing red and black balls into cells that contains the problem as a special case.

32. $N = \binom{52}{13}$ $\quad N_{fav} = \binom{13}{a}\binom{13}{b}\binom{13}{c}\binom{13}{d}$

$$\Rightarrow P(a,b,c,d) = \frac{\binom{13}{a}\binom{13}{b}\binom{13}{c}\binom{13}{d}}{\binom{52}{13}}$$

33. Using the result of problem 32, find the probability that some player receives a, another b, a third c, and the last d spades if (a) $a = 5$, $b = 4$, $c = 3$, $d = 1$; (b) $a = b = c = 4$, $d = 1$; (c) $a = b = 4$, $c = 3$, $d = 2$. Note that the three cases are essentially different.

a) $$P(a,b,c,d) = P(5,4,3,2,1)\cdot 4! = \frac{\binom{13}{5}\binom{13}{4}\binom{13}{2}\binom{13}{1}}{\binom{52}{13}}\cdot 4!$$

b) $$P(a,b,c,d) = 4\cdot P(4,4,4,1) = \frac{\binom{13}{4}^3\binom{13}{1}}{\binom{52}{13}}$$

c) $$P(a,b,c,d) = \binom{4}{2}\cdot P(4,4,3,2) = 12\cdot\frac{\binom{13}{4}^2\binom{13}{3}\binom{13}{2}}{\binom{52}{13}}$$

35. *Distribution of aces among r bridge cards.* Calculate the probabilities $p_0(r), p_1(r), \ldots, p_4(r)$ that among r bridge cards drawn at random there are $0, 1, \ldots, 4$ aces, respectively. Verify that $p_0(r) = p_4(52-r)$.

35.

$$P_k(r) = \frac{\binom{4}{k}\binom{48}{r-k}}{\binom{52}{r}}$$

$\Rightarrow$ Para $P_0(r) = P_4(52-r)$

$$\frac{\cancel{\binom{4}{0}}^{1}\binom{48}{r}}{\binom{52}{r}} = \frac{\cancel{\binom{4}{4}}^{1}\binom{48}{48-r}}{\binom{52}{52-r}}$$

$$\frac{\binom{48}{r}}{\binom{52}{r}} = \frac{\binom{48}{48-r}}{\binom{52}{52-r}}$$

$$\frac{\frac{\cancel{48!}}{r!(48-r)!}}{\frac{\cancel{52!}}{\cancel{r!}(\cancel{52-r})!}} = \frac{\frac{\cancel{48!}}{(48-r)!(48-48+r)!}}{\frac{\cancel{52!}}{(\cancel{52-r})!\cancel{(r)!}}}$$

$$r!(48-r)! = (48-r)!(r)!$$

$$1 = 1$$

Si son iguales

$$P_0(r) = P_4(52-r)$$

36. *Continuation: waiting times.* If the cards are drawn one by one, find the probabilities $f_1(r), \ldots, f_4(r)$ that the first, ..., fourth ace turns up at the rth trial. *Guess* at the *medians* of the waiting times for the first, ..., fourth ace and then calculate them.

36. Esto es una distribución hipergeométrica negativa

$$\Rightarrow f_K(r) = \frac{\binom{4}{K-1}\binom{48}{(r-1)-(K-1)}}{\binom{52}{r-1}} \cdot \frac{4-(K-1)}{52-(r-1)}$$

37. Find the probability that each of two hands contains exactly k aces if the two hands are composed of r bridge cards each, and are drawn (*a*) from the same deck, (*b*) from two decks. Show that when $r = 13$ the probability in part (*a*) is the probability that two preassigned bridge players receive exactly k aces each.

37.

a) $P = P(\text{mano 1 con } K) \cdot P(\text{mano2 con } K \mid \text{Mano 1 con } K)$

$$P = \frac{\binom{4}{K}\binom{48}{r-K}}{\binom{52}{r}} \times \frac{\binom{4-K}{K}\binom{48-(r-K)}{r-K}}{\binom{52-r}{r}}$$

b) $$P = \left[\frac{\binom{4}{K}\binom{48}{r-K}}{\binom{52}{r}}\right]^2$$

39. If r_1 indistinguishable things of one kind and r_2 indistinguishable things of a second kind are placed into n cells, find the number of distinguishable arrangements.

Supongamos que en vez de tomar objetos, seleccionamos celdas =>

$$N = \frac{n!}{(n-r_1-r_2)!} \quad N_{fav} = \binom{n}{r_1}\cdot\binom{n-r_1}{r_2} = \frac{n!\cdot\cancel{(n-r_1)!}}{r_1!\,\cancel{(n-r_1)!}\,(n-r_1-r_2)!\,r_2!} = \frac{n!}{r_1!\,(n-r_1-r_2)!\,r_2!}$$

$$\Rightarrow P_N = \frac{\frac{n!}{r_1!\,\cancel{(n-r_1-r_2)!}\,r_2!}}{\frac{n!}{\cancel{(n-r_1-r_2)!}}} = \frac{n!}{r_1!\,n!\,r_2!} = \frac{1}{r_1!\cdot r_2!}$$

40. If r_1 dice and r_2 coins are thrown, how many results can be distinguished?

Suponiendo que r_1 tiene 6 caras y r_2 dos:

$$N = 6^{r_1}\cdot 2^{r_2} \quad N_f = \binom{r_1+r_2}{r_1}$$

$$\Rightarrow P_N = \frac{\frac{r_1+r_2}{r_1}}{6^{r_1}\cdot 2^{r_2}} = \frac{r_1+r_2}{r_1\cdot 6^{r_1}\cdot 2^{r_2}}$$

41. In how many different distinguishable ways can r_1 white, r_2 black, and r_3 red balls be arranged?

$$\mathcal{X} = \frac{N!}{r_1!\cdot r_2!\cdot r_3!}$$

42. Find the probability that in a random arrangement of 52 bridge cards no two aces are adjacent.

$$N = \binom{52}{4} \quad N_f = \binom{49}{4} \qquad P_N = \frac{\binom{49}{4}}{\binom{52}{4}}$$

44. *Birthdays.* Find the probabilities for the various configurations of the birthdays of 22 people.

$n = 365$ días $\quad r = 22$ personas $\Rightarrow N = n^r = 365^{22}$

$$\Rightarrow N_f = \frac{n!}{(n-(r-1))!} = \frac{n!}{(n-r+1)!}$$

$$\therefore P_N = \frac{n!}{(n-r+1)!\,365^{22}}$$

45. Find the probability for a *poker* hand to be a (*a*) royal flush (ten, jack, queen, king, ace in a single suit); (*b*) four of a kind (four cards of equal face values); (*c*) full house (one pair and one triple of cards with equal face values); (*d*) straight (five cards in sequence regardless of suit); (*e*) three of a kind (three equal face values plus two extra cards); (*f*) two pairs (two pairs of equal face values plus one other card); (*g*) one pair (one pair of equal face values plus three different cards).

$N = \binom{52}{5}$

a) $P = \dfrac{4}{\binom{52}{5}}$

b) $P = \dfrac{13 \cdot 48}{\binom{52}{5}}$

c) $P = \dfrac{\binom{13}{1}\binom{4}{3}\binom{12}{1}\binom{4}{2}}{\binom{52}{5}}$

d) $P = \dfrac{10 \cdot 4^5 - 40}{\binom{52}{5}}$

e) $P = \dfrac{\binom{13}{1}\binom{4}{3} \cdot \binom{12}{2}\binom{4}{1}^2}{\binom{54}{5}}$

f) $P = \dfrac{\binom{13}{2}\binom{4}{2}^2 \binom{11}{1}\binom{4}{1}}{\binom{52}{5}}$

g) $P = \dfrac{\binom{13}{1}\binom{4}{2}\binom{12}{3}\binom{4}{1}^3}{\binom{52}{5}}$